

There are a number of texts on discrete-time signals and systems. We mention as examples the books by McGillem and Cooper (1984), Oppenheim and Willsky (1983), and Siebert (1986). Linear constant-coefficient difference equations are treated in depth in the books by Hildebrand (1952) and Levy and Lessman (1961).

The last topic in this chapter, on correlation of discrete-time signals, plays an important role in digital signal processing, especially in applications dealing with digital communications, radar detection and estimation, sonar, and geophysics. In our treatment of correlation sequences, we avoided the use of statistical concepts. Correlation is simply defined as a mathematical operation between two sequences, which produces another sequence, called either the *crosscorrelation sequence* when the two sequences are different, or the *autocorrelation sequence* when the two sequences are identical.

In practical applications in which correlation is used, one (or both) of the sequences is (are) contaminated by noise and, perhaps, by other forms of interference. In such a case, the noisy sequence is called a *random sequence* and is characterized in statistical terms. The corresponding correlation sequence becomes a function of the statistical characteristics of the noise and any other interference.

Supplementary reading on probabilistic and statistical concepts dealing with correlation can be found in the books by Davenport (1970), Helstrom (1990), Peebles (1987), and Stark and Woods (1994).

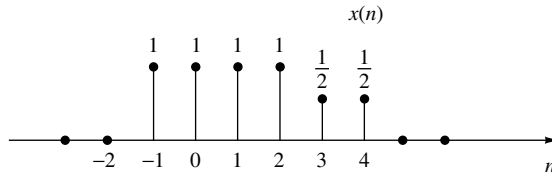
Problems

- 1 A discrete-time signal $x(n)$ is defined as

$$x(n) = \begin{cases} 1 + \frac{n}{3}, & -3 \leq n \leq -1 \\ 1, & 0 \leq n \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

- (a) Determine its values and sketch the signal $x(n)$.
 - (b) Sketch the signals that result if we:
 - 1. First fold $x(n)$ and then delay the resulting signal by four samples.
 - 2. First delay $x(n)$ by four samples and then fold the resulting signal.
 - (c) Sketch the signal $x(-n + 4)$.
 - (d) Compare the results in parts (b) and (c) and derive a rule for obtaining the signal $x(-n + k)$ from $x(n)$.
 - (e) Can you express the signal $x(n)$ in terms of signals $\delta(n)$ and $u(n)$?
- 2 A discrete-time signal $x(n)$ is shown in Fig. P2. Sketch and label carefully each of the following signals.

Figure P2



- (a) $x(n-2)$ (b) $x(4-n)$ (c) $x(n+2)$ (d) $x(n)u(2-n)$ (e) $x(n-1)\delta(n-3)$
 (f) $x(n^2)$ (g) even part of $x(n)$ (h) odd part of $x(n)$

3 Show that

(a) $\delta(n) = u(n) - u(n-1)$

(b) $u(n) = \sum_{k=-\infty}^n \delta(k) = \sum_{k=0}^{\infty} \delta(n-k)$

4 Show that any signal can be decomposed into an even and an odd component. Is the decomposition unique? Illustrate your arguments using the signal

$$x(n) = \{2, 3, 4, 5, 6\}$$

↑

- 5 Show that the energy (power) of a real-valued energy (power) signal is equal to the sum of the energies (powers) of its even and odd components.
 6 Consider the system

$$y(n) = \mathcal{T}[x(n)] = x(n^2)$$

- (a) Determine if the system is time invariant.
 (b) To clarify the result in part (a) assume that the signal

$$x(n) = \begin{cases} 1, & 0 \leq n \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

is applied into the system.

- (1) Sketch the signal $x(n)$.
 (2) Determine and sketch the signal $y(n) = \mathcal{T}[x(n)]$.
 (3) Sketch the signal $y'_2(n) = y(n-2)$.
 (4) Determine and sketch the signal $x_2(n) = x(n-2)$.
 (5) Determine and sketch the signal $y_2(n) = \mathcal{T}[x_2(n)]$.
 (6) Compare the signals $y_2(n)$ and $y(n-2)$. What is your conclusion?
 (c) Repeat part (b) for the system

$$y(n) = x(n) - x(n-1)$$

Can you use this result to make any statement about the time invariance of this system? Why?

- (d) Repeat parts (b) and (c) for the system

$$y(n) = \mathcal{T}[x(n)] = nx(n)$$

7 A discrete-time system can be

- (1) Static or dynamic
- (2) Linear or nonlinear
- (3) Time invariant or time varying
- (4) Causal or noncausal
- (5) Stable or unstable

Examine the following systems with respect to the properties above.

- (a) $y(n) = \cos[x(n)]$
- (b) $y(n) = \sum_{k=-\infty}^{n+1} x(k)$
- (c) $y(n) = x(n) \cos(\omega_0 n)$
- (d) $y(n) = x(-n + 2)$
- (e) $y(n) = \text{Trun}[x(n)]$, where $\text{Trun}[x(n)]$ denotes the integer part of $x(n)$, obtained by truncation
- (f) $y(n) = \text{Round}[x(n)]$, where $\text{Round}[x(n)]$ denotes the integer part of $x(n)$ obtained by rounding

Remark: The systems in parts (e) and (f) are quantizers that perform truncation and rounding, respectively.

- (g) $y(n) = |x(n)|$
 - (h) $y(n) = x(n)u(n)$
 - (i) $y(n) = x(n) + nx(n + 1)$
 - (j) $y(n) = x(2n)$
 - (k) $y(n) = \begin{cases} x(n), & \text{if } x(n) \geq 0 \\ 0, & \text{if } x(n) < 0 \end{cases}$
 - (l) $y(n) = x(-n)$
 - (m) $y(n) = \text{sign}[x(n)]$
 - (n) The ideal sampling system with input $x_a(t)$ and output $x(n) = x_a(nT)$, $-\infty < n < \infty$
- 8** Two discrete-time systems $\mathcal{T}_1$ and $\mathcal{T}_2$ are connected in cascade to form a new system $\mathcal{T}$ as shown in Fig. P8. Prove or disprove the following statements.
- (a) If $\mathcal{T}_1$ and $\mathcal{T}_2$ are linear, then $\mathcal{T}$ is linear (i.e., the cascade connection of two linear systems is linear).
 - (b) If $\mathcal{T}_1$ and $\mathcal{T}_2$ are time invariant, then $\mathcal{T}$ is time invariant.
 - (c) If $\mathcal{T}_1$ and $\mathcal{T}_2$ are causal, then $\mathcal{T}$ is causal.
 - (d) If $\mathcal{T}_1$ and $\mathcal{T}_2$ are linear and time invariant, the same holds for $\mathcal{T}$.
 - (e) If $\mathcal{T}_1$ and $\mathcal{T}_2$ are linear and time invariant, then interchanging their order does not change the system $\mathcal{T}$.
 - (f) As in part (e) except that $\mathcal{T}_1, \mathcal{T}_2$ are now time varying. (*Hint:* Use an example.)
 - (g) If $\mathcal{T}_1$ and $\mathcal{T}_2$ are nonlinear, then $\mathcal{T}$ is nonlinear.
 - (h) If $\mathcal{T}_1$ and $\mathcal{T}_2$ are stable, then $\mathcal{T}$ is stable.
 - (i) Show by an example that the inverses of parts (c) and (h) do not hold in general.

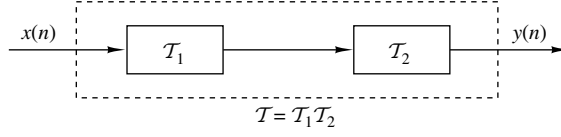


Figure P8

- 9 Let $\mathcal{T}$ be an LTI, relaxed, and BIBO stable system with input $x(n)$ and output $y(n)$. Show that:
- (a) If $x(n)$ is periodic with period N [i.e., $x(n) = x(n + N)$ for all $n \geq 0$], the output $y(n)$ tends to a periodic signal with the same period.
 - (b) If $x(n)$ is bounded and tends to a constant, the output will also tend to a constant.
 - (c) If $x(n)$ is an energy signal, the output $y(n)$ will also be an energy signal.
- 10 The following input–output pairs have been observed during the operation of a time-invariant system:

$$\begin{aligned} x_1(n) = \{1, 0, 2\} &\xleftrightarrow{\mathcal{T}} y_1(n) = \{0, 1, 2\} \\ &\quad \uparrow \qquad \qquad \uparrow \\ x_2(n) = \{0, 0, 3\} &\xleftrightarrow{\mathcal{T}} y_2(n) = \{0, 1, 0, 2\} \\ &\quad \uparrow \qquad \qquad \uparrow \\ x_3(n) = \{0, 0, 0, 1\} &\xleftrightarrow{\mathcal{T}} y_3(n) = \{1, 2, 1\} \\ &\quad \uparrow \qquad \qquad \uparrow \end{aligned}$$

Can you draw any conclusions regarding the linearity of the system. What is the impulse response of the system?

- 11 The following input–output pairs have been observed during the operation of a linear system:

$$\begin{aligned} x_1(n) = \{-1, 2, 1\} &\xleftrightarrow{\mathcal{T}} y_1(n) = \{1, 2, -1, 0, 1\} \\ &\quad \uparrow \qquad \qquad \uparrow \\ x_2(n) = \{1, -1, -1\} &\xleftrightarrow{\mathcal{T}} y_2(n) = \{-1, 1, 0, 2\} \\ &\quad \uparrow \qquad \qquad \uparrow \\ x_3(n) = \{0, 1, 1\} &\xleftrightarrow{\mathcal{T}} y_3(n) = \{1, 2, 1\} \\ &\quad \uparrow \qquad \qquad \uparrow \end{aligned}$$

Can you draw any conclusions about the time invariance of this system?

- 12 The only available information about a system consists of N input–output pairs, of signals $y_i(n) = \mathcal{T}[x_i(n)]$, $i = 1, 2, \dots, N$.
- (a) What is the class of input signals for which we can determine the output, using the information above, if the system is known to be linear?
 - (b) The same as above, if the system is known to be time invariant.
- 13 Show that the necessary and sufficient condition for a relaxed LTI system to be BIBO stable is

$$\sum_{n=-\infty}^{\infty} |h(n)| \leq M_h < \infty$$

for some constant M_h .

14 Show that:

(a) A relaxed linear system is causal if and only if for any input $x(n)$ such that

$$x(n) = 0 \text{ for } n < n_0 \Rightarrow y(n) = 0 \text{ for } n < n_0$$

(b) A relaxed LTI system is causal if and only if

$$h(n) = 0 \text{ for } n < 0$$

15

(a) Show that for any real or complex constant a , and any finite integer numbers M and N , we have

$$\sum_{n=0}^N a^n = \begin{cases} \frac{a^{N+1} - 1}{a - 1}, & \text{if } a \neq 1 \\ N + 1, & \text{if } a = 1 \end{cases}$$

(b) Show that if $|a| < 1$, then

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1 - a}$$

16 (a) If $y(n) = x(n) * h(n)$, show that $\sum_y = \sum_x \sum_h$, where $\sum_x = \sum_{n=-\infty}^{\infty} x(n)$.

(b) Compute the convolution $y(n) = x(n) * h(n)$ of the following signals and check the correctness of the results by using the test in (a).

(1) $x(n) = \{1, 2, 4\}, h(n) = \{1, 1, 1, 1\}$

(2) $x(n) = \{1, 2, -1\}, h(n) = x(n)$

(3) $x(n) = \{0, 1, -2, 3, -4\}, h(n) = \{\frac{1}{2}, \frac{1}{2}, 1, \frac{1}{2}\}$

(4) $x(n) = \{1, 2, 3, 4, 5\}, h(n) = \{1\}$

(5) $x(n) = \{\underset{\uparrow}{1}, -2, 3\}, h(n) = \{\underset{\uparrow}{0}, 0, 1, 1, 1, 1\}$

(6) $x(n) = \{\underset{\uparrow}{0}, 0, 1, 1, 1, 1\}, h(n) = \{1, \underset{\uparrow}{-2}, 3\}$

(7) $x(n) = \{\underset{\uparrow}{0}, 1, 4, -3\}, h(n) = \{\underset{\uparrow}{1}, 0, -1, -1\}$

(8) $x(n) = \{\underset{\uparrow}{1}, 1, 2\}, h(n) = u(n)$

(9) $x(n) = \{1, 1, \underset{\uparrow}{0}, 1, 1\}, h(n) = \{1, -2, -3, \underset{\uparrow}{4}\}$

(10) $x(n) = \{1, 2, \underset{\uparrow}{0}, 2, 1\} h(n) = x(n)$

(11) $x(n) = (\frac{1}{2})^n u(n), h(n) = (\frac{1}{4})^n u(n)$

- 17** Compute and plot the convolutions $x(n) * h(n)$ and $h(n) * x(n)$ for the pairs of signals shown in Fig. P17.

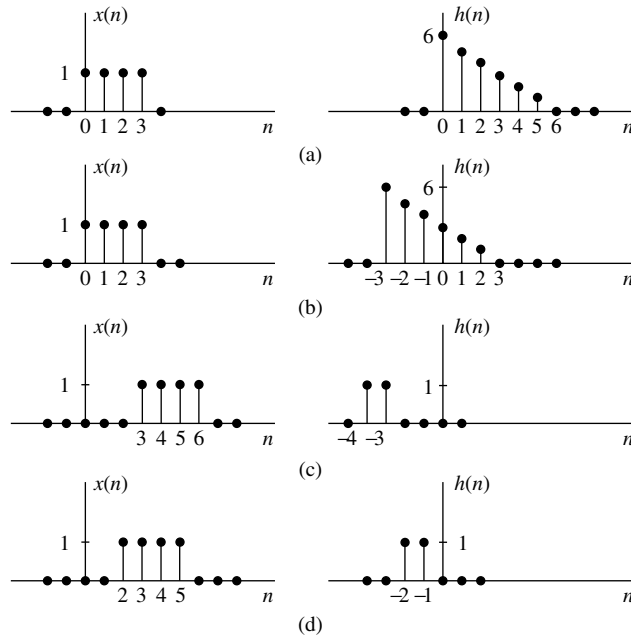


Figure P17

- 18** Determine and sketch the convolution $y(n)$ of the signals

$$x(n) = \begin{cases} \frac{1}{3}n, & 0 \leq n \leq 6 \\ 0, & \text{elsewhere} \end{cases}$$

$$h(n) = \begin{cases} 1, & -2 \leq n \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

- (a) Graphically
(b) Analytically

- 19** Compute the convolution $y(n)$ of the signals

$$x(n) = \begin{cases} \alpha^n, & -3 \leq n \leq 5 \\ 0, & \text{elsewhere} \end{cases}$$

$$h(n) = \begin{cases} 1, & 0 \leq n \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

- 20** Consider the following three operations.

- (a) Multiply the integer numbers: 131 and 122.
(b) Compute the convolution of signals: $\{1, 3, 1\} * \{1, 2, 2\}$.
(c) Multiply the polynomials: $1 + 3z + z^2$ and $1 + 2z + 2z^2$.
(d) Repeat part (a) for the numbers 1.31 and 12.2.
(e) Comment on your results.

21 Compute the convolution $y(n) = x(n) * h(n)$ of the following pairs of signals.

(a) $x(n) = a^n u(n)$, $h(n) = b^n u(n)$ when $a \neq b$ and when $a = b$

(b) $x(n) = \begin{cases} 1, & n = -2, 0, 1 \\ 2, & n = -1 \\ 0, & \text{elsewhere} \end{cases}$ $h(n) = \delta(n) - \delta(n-1) + \delta(n-4) + \delta(n-5)$

(c) $x(n) = u(n+1) - u(n-4) - \delta(n-5)$; $h(n) = [u(n+2) - u(n-3)] \cdot (3 - |n|)$

(d) $x(n) = u(n) - u(n-5)$; $h(n) = u(n-2) - u(n-8) + u(n-11) - u(n-17)$

22 Let $x(n]$ be the input signal to a discrete-time filter with impulse response $h_i(n)$ and let $y_i(n)$ be the corresponding output.

(a) Compute and sketch $x(n)$ and $y_i(n)$ in the following cases, using the same scale in all figures.

$$x(n) = \{1, 4, 2, 3, 5, 3, 3, 4, 5, 7, 6, 9\}$$

$$h_1(n) = \{1, 1\}$$

$$h_2(n) = \{1, 2, 1\}$$

$$h_3(n) = \{\frac{1}{2}, \frac{1}{2}\}$$

$$h_4(n) = \{\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\}$$

$$h_5(n) = \{\frac{1}{4}, -\frac{1}{2}, \frac{1}{4}\}$$

Sketch $x(n)$, $y_1(n)$, $y_2(n)$ on one graph and $x(n)$, $y_3(n)$, $y_4(n)$, $y_5(n)$ on another graph

(b) What is the difference between $y_1(n)$ and $y_2(n)$, and between $y_3(n)$ and $y_4(n)$?

(c) Comment on the smoothness of $y_2(n)$ and $y_4(n)$. Which factors affect the smoothness?

(d) Compare $y_4(n)$ with $y_5(n)$. What is the difference? Can you explain it?

(e) Let $h_6(n) = \{\frac{1}{2}, -\frac{1}{2}\}$. Compute $y_6(n)$. Sketch $x(n)$, $y_2(n)$, and $y_6(n)$ on the same figure and comment on the results.

23 Express the output $y(n)$ of a linear time-invariant system with impulse response $h(n)$ in terms of its step response $s(n) = h(n) * u(n)$ and the input $x(n)$.

24 The discrete-time system

$$y(n) = ny(n-1) + x(n), \quad n \geq 0$$

is at rest [i.e., $y(-1) = 0$]. Check if the system is linear time invariant and BIBO stable.

25 Consider the signal $y(n) = a^n u(n)$, $0 < a < 1$.

- (a) Show that any sequence $x(n)$ can be decomposed as

$$x(n) = \sum_{k=-\infty}^{\infty} c_k \gamma(n-k)$$

and express c_k in terms of $x(n)$.

- (b) Use the properties of linearity and time invariance to express the output $y(n) = \mathcal{T}[x(n)]$ in terms of the input $x(n)$ and the signal $g(n) = \mathcal{T}[\gamma(n)]$, where $\mathcal{T}[\cdot]$ is an LTI system.
- (c) Express the impulse response $h(n) = \mathcal{T}[\delta(n)]$ in terms of $g(n)$.
- 26 Determine the zero-input response of the system described by the second-order difference equation

$$x(n) - 3y(n-1) - 4y(n-2) = 0$$

- 27 Determine the particular solution of the difference equation

$$y(n) = \frac{5}{6}y(n-1) - \frac{1}{6}y(n-2) + x(n)$$

when the forcing function is $x(n) = 2^n u(n)$.

- 28 In Example 4.8, equation (4.30), separate the output sequence $y(n)$ into the transient response and the steady-state response. Plot these two responses for $a_1 = -0.9$.
- 29 Determine the impulse response for the cascade of two linear time-invariant systems having impulse responses.

$$h_1(n) = a^n[u(n) - u(n-N)] \text{ and } h_2(n) = [u(n) - u(n-M)]$$

- 30 Determine the response $y(n)$, $n \geq 0$, of the system described by the second-order difference equation

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$$

to the input $x(n) = 4^n u(n)$.

- 31 Determine the impulse response of the following causal system:

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$$

- 32 Let $x(n)$, $N_1 \leq n \leq N_2$ and $h(n)$, $M_1 \leq n \leq M_2$ be two finite-duration signals.

- (a) Determine the range $L_1 \leq n \leq L_2$ of their convolution, in terms of N_1 , N_2 , M_1 and M_2 .
- (b) Determine the limits of the cases of partial overlap from the left, full overlap, and partial overlap from the right. For convenience, assume that $h(n)$ has shorter duration than $x(n)$.
- (c) Illustrate the validity of your results by computing the convolution of the signals

$$x(n) = \begin{cases} 1, & -2 \leq n \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

$$h(n) = \begin{cases} 2, & -1 \leq n \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

- 33** Determine the impulse response and the unit step response of the systems described by the difference equation

(a) $y(n] = 0.6y(n - 1) - 0.08y(n - 2) + x(n]$

(b) $y(n] = 0.7y(n - 1) - 0.1y(n - 2) + 2x(n] - x(n - 2)$

- 34** Consider a system with impulse response

$$h(n] = \begin{cases} (\frac{1}{2})^n, & 0 \leq n \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

Determine the input $x(n]$ for $0 \leq n \leq 8$ that will generate the output sequence

$$y(n] = \{1, 2, 2.5, 3, 3, 3, 2, 1, 0, \dots\}$$

- 35** Consider the interconnection of LTI systems as shown in Fig. P35.

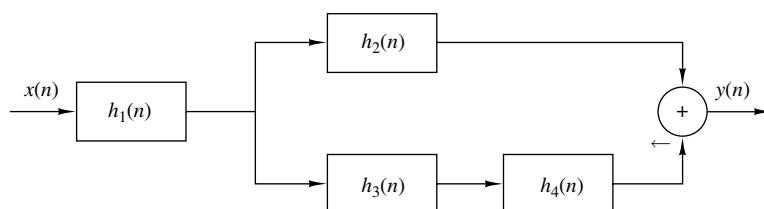


Figure P35

- (a) Express the overall impulse response in terms of $h_1(n]$, $h_2(n]$, $h_3(n]$, and $h_4(n]$.

- (b) Determine $h(n]$ when

$$h_1(n] = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\}$$

$$h_2(n] = h_3(n] = (n + 1)u(n]$$

$$h_4(n] = \delta(n - 2)$$

- (c) Determine the response of the system in part (b) if

$$x(n] = \delta(n + 2) + 3\delta(n - 1) - 4\delta(n - 3)$$

- 36** Consider the system in Fig. P36 with $h(n] = a^n u(n]$, $-1 < a < 1$. Determine the response $y(n]$ of the system to the excitation

$$x(n] = u(n + 5) - u(n - 10)$$

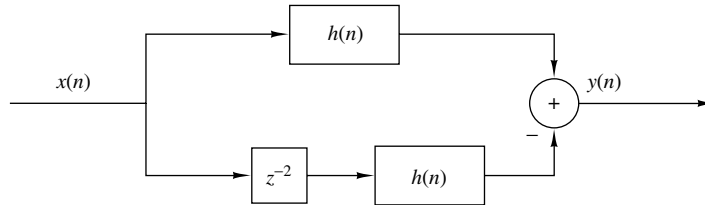


Figure P36

- 37** Compute and sketch the step response of the system

$$y(n) = \frac{1}{M} \sum_{k=0}^{M-1} x(n-k)$$

- 38** Determine the range of values of the parameter a for which the linear time-invariant system with impulse response

$$h(n) = \begin{cases} a^n, & n \geq 0, n \text{ even} \\ 0, & \text{otherwise} \end{cases}$$

is stable.

- 39** Determine the response of the system with impulse response

$$h(n) = a^n u(n)$$

to the input signal

$$x(n) = u(n) - u(n-10)$$

(Hint: The solution can be obtained easily and quickly by applying the linearity and time-invariance properties to the result in Example 3.5.)

- 40** Determine the response of the (relaxed) system characterized by the impulse response

$$h(n) = \left(\frac{1}{2}\right)^n u(n)$$

to the input signal

$$x(n) = \begin{cases} 1, & 0 \leq n < 10 \\ 0, & \text{otherwise} \end{cases}$$

- 41** Determine the response of the (relaxed) system characterized by the impulse response

$$h(n) = \left(\frac{1}{2}\right)^n u(n)$$

to the input signals

(a) $x(n) = 2^n u(n)$

(b) $x(n) = u(-n)$

- 42** Three systems with impulse responses $h_1(n) = \delta(n) - \delta(n - 1)$, $h_2(n) = h(n)$, and $h_3(n) = u(n)$, are connected in cascade.

- (a) What is the impulse response, $h_c(n)$, of the overall system?
 (b) Does the order of the interconnection affect the overall system?

- 43** (a) Prove and explain graphically the difference between the relations

$$x(n)\delta(n - n_0) = x(n_0)\delta(n - n_0) \quad \text{and} \quad x(n) * \delta(n - n_0) = x(n - n_0)$$

- (b) Show that a discrete-time system, which is described by a convolution summation, is LTI and relaxed,

- (c) What is the impulse response of the system described by $y(n) = x(n - n_0)$?

- 44** Two signals $s(n)$ and $v(n)$ are related through the following difference equations.

$$s(n) + a_1s(n - 1) + \cdots + a_Ns(n - N) = b_0v(n)$$

Design the block diagram realization of:

- (a) The system that generates $s(n)$ when excited by $v(n)$.
 (b) The system that generates $v(n)$ when excited by $s(n)$.
 (c) What is the impulse response of the cascade interconnection of systems in parts (a) and (b)?

- 45** Compute the zero-state response of the system described by the difference equation

$$y(n) + \frac{1}{2}y(n - 1) = x(n) + 2x(n - 2)$$

to the input

$$x(n) = \{1, 2, \underset{\uparrow}{3}, 4, 2, 1\}$$

by solving the difference equation recursively.

- 46** Determine the direct form II realization for each of the following LTI systems:

- (a) $2y(n) + y(n - 1) - 4y(n - 3) = x(n) + 3x(n - 5)$
 (b) $y(n) = x(n) - x(n - 1) + 2x(n - 2) - 3x(n - 4)$

- 47** Consider the discrete-time system shown in Fig. P47.

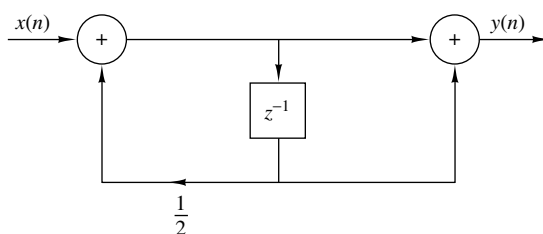


Figure P47

- (a) Compute the 10 first samples of its impulse response.
 (b) Find the input–output relation.
 (c) Apply the input $x(n] = \{1, 1, 1, \dots\}$ and compute the first 10 samples of the output.
 (d) Compute the first 10 samples of the output for the input given in part (c) by using convolution.
 (e) Is the system causal? Is it stable?
- 48 Consider the system described by the difference equation

$$y(n] = ay(n - 1] + bx(n]$$

- (a) Determine b in terms of a so that

$$\sum_{n=-\infty}^{\infty} h(n] = 1$$

- (b) Compute the zero-state step response $s(n]$ of the system and choose b so that $s(\infty) = 1$.
 (c) Compare the values of b obtained in parts (a) and (b). What did you notice?
- 49 A discrete-time system is realized by the structure shown in Fig. P49.
 (a) Determine the impulse response.
 (b) Determine a realization for its inverse system, that is, the system which produces $x(n]$ as an output when $y(n]$ is used as an input.

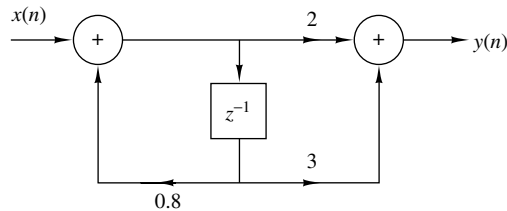


Figure P49

- 50 Consider the discrete-time system shown in Fig. P50.

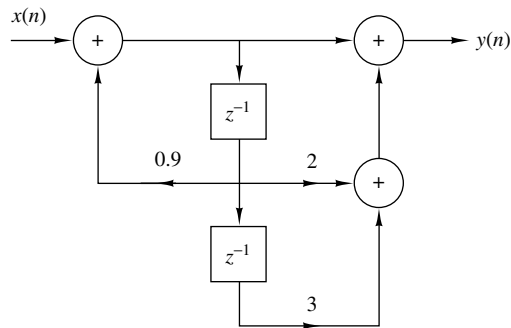


Figure P50

- (a) Compute the first six values of the impulse response of the system.
 (b) Compute the first six values of the zero-state step response of the system.
 (c) Determine an analytical expression for the impulse response of the system.
- 51 Determine and sketch the impulse response of the following systems for $n = 0, 1, \dots, 9$.

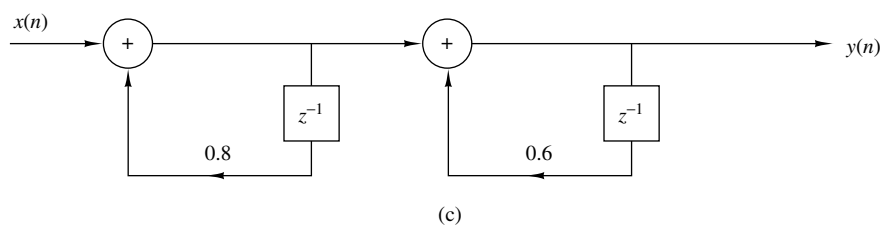
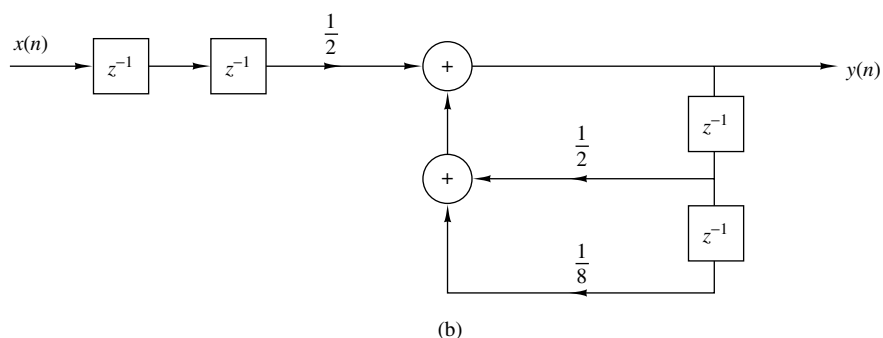
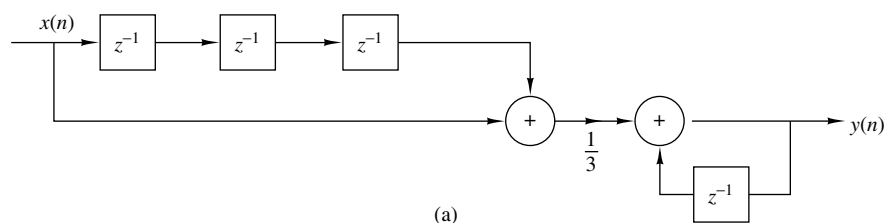


Figure P51

- (a) Fig. P51(a).
 (b) Fig. P51(b).
 (c) Fig. P51(c).
 (d) Classify the systems above as FIR or IIR.
 (e) Find an explicit expression for the impulse response of the system in part (c).

52 Consider the systems shown in Fig. P52.

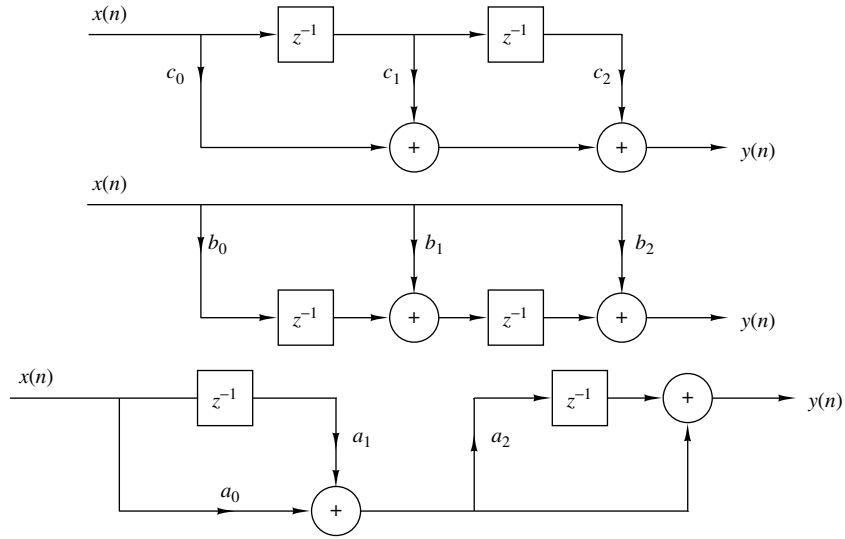


Figure P52

- (a) Determine and sketch their impulse responses $h_1(n)$, $h_2(n)$, and $h_3(n)$.
 (b) Is it possible to choose the coefficients of these systems in such a way that

$$h_1(n) = h_2(n) = h_3(n)$$

53 Consider the system shown in Fig. P53.

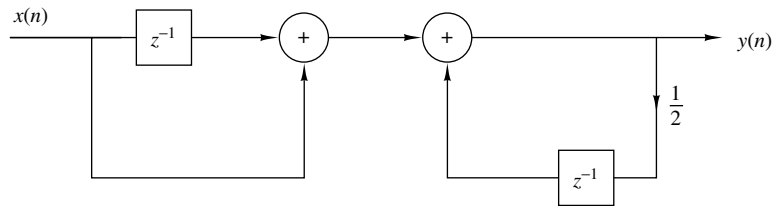


Figure P53

- (a) Determine its impulse response $h(n)$.
 (b) Show that $h(n)$ is equal to the convolution of the following signals:

$$h_1(n) = \delta(n) + \delta(n - 1)$$

$$h_2(n) = \left(\frac{1}{2}\right)^n u(n)$$

54 Compute and sketch the convolution $y_i(n)$ and correlation $r_i(n)$ sequences for the following pair of signals and comment on the results obtained.

(a) $x_1(n) = \{1, 2, 4\}$ $h_1(n) = \{1, 1, 1, 1, 1\}$

(b) $x_2(n) = \{0, 1, -2, 3, -4\}$ $h_2(n) = \{\frac{1}{2}, 1, 2, 1, \frac{1}{2}\}$